ACS Catalysis

pubs.acs.org/acscatalysis Research Article

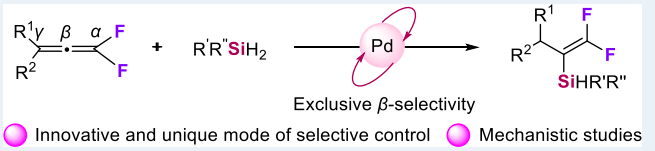
Chemo- and Regioselective Palladium-Catalyzed Internal Hydrosilylation of *gem*-Difluoroallenes to Access β-Fluorinated Vinylsilanes

Xin-Long Yu, An-Jiu Wen, Zheng Xu, Samuel Couve-Bonnaire,* Li-Wen Xu,* and Jean-Philippe Bouillon*

Cite This: *ACS Catal.* 2025, 15, 13747−13756 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Hydrosilylation represents one of the most important transformations in organosilicon chemistry and homogeneous catalysis. While terminal alkene hydrosilylation has been extensively studied, catalytic hydrosilylation of internal alkenes remains significantly underdeveloped. In this work, we report a highly efficient palladium-catalyzed internal hydrosilylation of *gem*-difluoroallenes, providing straightforward access to valuable *gem*-difluorovinylsilanes. The reaction proceeds with excellent regio- and chemoselectivity under mild conditions, demonstrating remarkable potential for late-stage functionalization applications. Through comprehensive Density Functional Theory (DFT) studies, we have elucidated the critical role of the DPEphos ligand in facilitating the internal hydrosilylation of *gem*-difluoroallenes, offering mechanistic insights into the control of reaction selectivity. This novel approach to selective *gem*-difluoroallene transformation enables the synthesis of previously inaccessible *gem*-difluorovinylsilanes, which serve as versatile building blocks for diverse chemical transformations.



KEYWORDS: *gem-difluoroallene, vinylsilane, palladium, catalysis, hydrosilylation*

INTRODUCTION

The fluorine atom is considered to significantly alter the physicochemical properties of drug molecules due to its high electronegativity and small size.[1] Additionally, the introduction of fluorine-containing groups could enhance the lipophilicity, metabolic stability, and membrane permeability of the molecules, thereby improving their efficacy and pharmacokinetic properties.[2] In recent years, *gem*-difluoroalkenes have attracted much attention in the field of medicinal chemistry[3] as a bioisostere of carbonyl group, which could effectively improve the metabolic stability of molecules and optimize the biological activity by modulating the electronic properties of the target binding sites.[4] For instance, the replacement of the carbonyl group in the artemisinin molecule by a *gem*-difluoroalkene moiety significantly enhanced the antimalarial activity, suggesting that its structure showed effects on the interaction of the molecule with the biological targets based on the adjustment of the electronic distribution and spatial conformation. Furthermore, the *gem*-difluoroalkene moiety exhibited unique electronic properties and could be employed in the regulation of the lipophilicity and metabolic stability of molecules, alongside the design of more selective targeted drugs making it a versatile module in drug design (Scheme 1A).[5]

Different from the typical allenes, *gem*-difluoroallenes possessed the unique backbone of 1,2-propadiene with two fluorine atoms at the terminal position. The electron-withdrawing effect of fluorine atoms showed significant effects on the reactivity of *gem*-difluoroallenes.[6] Moreover, due to its high electronegativity and strong C−F bond, the electrophilic α-carbon is more favorable to react with nucleophilic reagents. Although fluorine is not generally considered as a good leaving group, it can often be easily dissociated from *gem*-difluorinated olefins or activated alkyl fluoride groups under strongly nucleophilic or alkaline conditions.[7] Compared with the α- and γ-selective additions to *gem*-difluoroallene, β-addition reactions are relatively rare. In 2022, Shi's group reported the selective β-addition of thiols to *gem*-difluoroallenes by a rhodium catalyst, in which the β-adducts were obtained in fair to good yields (58−77%) with high chemo- and regioselectivity (up to 20:1 *rr*) leading to thioenolether with the difluoromethyl group in allylic position.[8] It was revealed by mechanistic studies that the reaction proceeded through a stable *Z*-rhodacyclic transition state, which enabled precise control over β-selectivity. However, to our knowledge, this is the only example reported achieving β-additions on *gem*-

Received: May 26, 2025
Revised: July 18, 2025
Accepted: July 18, 2025
Published: July 25, 2025

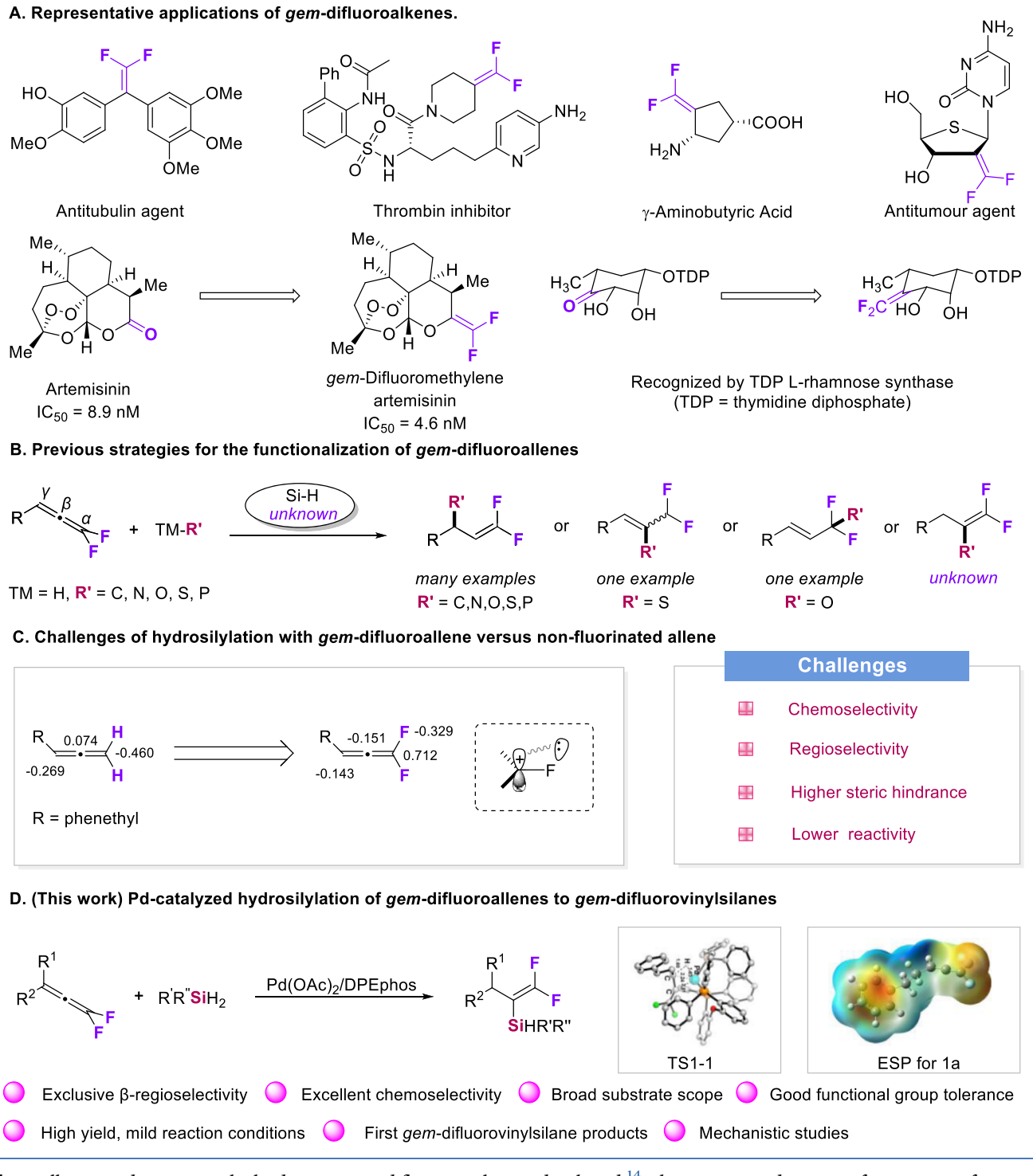


Downloaded via ZHEJIANG UNIV on January 30, 2026 at 03:35:40 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

ACS Publications © 2025 American Chemical Society

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Scheme 1. Regioselective Functionalization of *gem*-Difluoroallenes



difluoroallenes and no example leading to *gem*-difluorovinyl products has been reported so far.[9]

Although numerous studies have documented the addition reactions of various hydrogen-functionalized compounds (e.g., C−H, N−H, O−H, S−H, and P−H bonds) to *gem*-difluoroallenes (Scheme 1B),[10−13] the hydrosilylation of *gem*-difluoroallenes remains unexplored. Notably, existing methods predominantly afford α-functionalized products, whereas regioselective β-position addition to access *gem*-difluorovinyl compounds remains a significant challenge. Additionally, while hydrosilylation reactions of terminal alkenes are well-developed,[14] the corresponding transformations of internal alkenes are far less explored. This discrepancy arises from the higher reactivity and lower steric hindrance of terminal alkenes, which facilitate efficient coordination with metal catalysts and promote hydrosilylation. Moreover, controlling the regioselectivity of internal alkene hydrosilylation remains a critical challenge in this field (Scheme 1C).[15] To the best of our knowledge, only two examples of internal allene hydrosilylation have been reported,[16] in stark contrast to the well-established hydrosilylation of terminal allenes.[17]

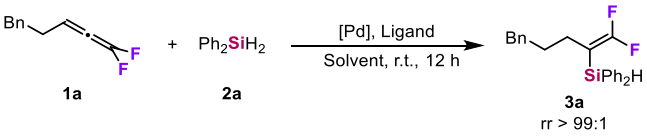
https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Herein, we report a palladium-catalyzed chemo- and regioselective hydrosilylation of *gem*-difluoroallenes, providing efficient access to *β*-difluorovinylsilanes under mild conditions (Scheme 1D). This method offers a new strategy for constructing difluorovinylsilanes and difluorovinyl-containing organic molecules.

RESULTS AND DISCUSSION

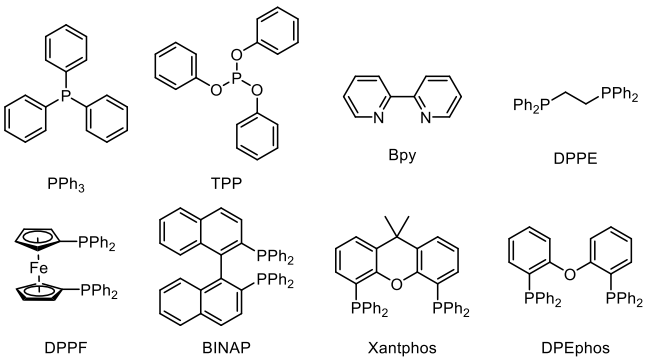
The *gem*-difluoroallene (**1a**) and diphenylsilane (**2a**) were used as model substrates to optimize the reaction. During the initial screening experiments, $Pd(OAc)_2$ was employed as the catalyst with toluene as the solvent at room temperature, in order to evaluate the performance of various ligands. Triphenylphosphine (PPh_3) produced trace amounts of the desired target product **3a** (Table 1, entry 1). Notably, the yield showed an increasing trend to 57% with the use of triphenylphosphite (TPP) as the ligand (Table 1, entry 2).

Table 1. Optimization of the Reaction Conditions[a]



entry	ligand	solvent	[Pd]	yield (%)[c]
1	PPh_3	toluene	$Pd(OAc)_2$	<10
2	TPP	toluene	$Pd(OAc)_2$	57
3	Bpy	toluene	$Pd(OAc)_2$	N.D.
4	DPPE	toluene	$Pd(OAc)_2$	N.D.
5	DPPF	toluene	$Pd(OAc)_2$	N.D.
6	BINAP	toluene	$Pd(OAc)_2$	18
7	Xantphos	toluene	$Pd(OAc)_2$	32
8	DPEphos	toluene	$Pd(OAc)_2$	64
9	DPEphos	DCM	$Pd(OAc)_2$	N.D.
10	DPEphos	hexane	$Pd(OAc)_2$	23
11	DPEphos	THF	$Pd(OAc)_2$	64
12	DPEphos	EtOAc	$Pd(OAc)_2$	71
13	DPEphos	1,4-dioxane	$Pd(OAc)_2$	98
14	DPEphos	1,4-dioxane	$Pd_2(dba)_3{\cdot}CHCl_3$	56
15	DPEphos	1,4-dioxane	$Pd(PPh_3)_4$	N.D.
16	DPEphos	1,4-dioxane	$PdBr_2$	N.D.
17	DPEphos[b]	1,4-dioxane	$Pd(OAc)_2$[b]	98 (88[d])

[a]



Substrate **1a** (0.2 mmol, 1.0 equiv), Ph_2SiH_2 (1.2 equiv, 0.24 mmol), palladium (10 mol %, 0.02 mmol), ligand (12 mol %, 0.024 mmol), solvent (2 mL) at room temperature under Ar for 12 h. [b]$Pd(OAc)_2$ (2 mol %, 0.004 mmol), DPEphos (3 mol %, 0.006 mmol). [c]Yield was determined by ^1H NMR analysis with CH_2Br_2 as the internal standard. [d]Isolated yield. N.D. = not determined.

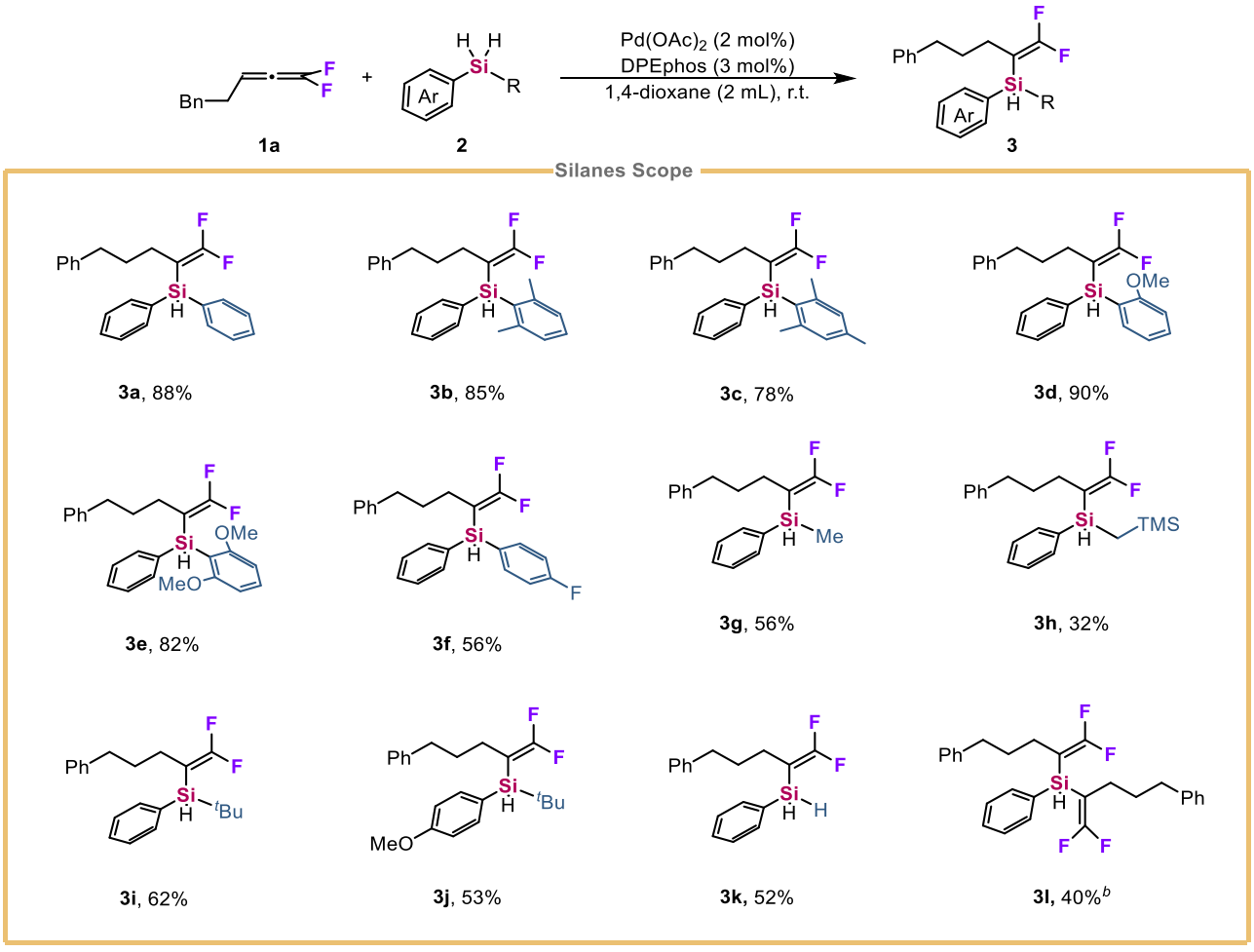
In contrast, nitrogen-based bidentate ligand (Bpy), bidentate phosphine ligand (DPPE), and ferrocenyl-based ligand (DPPF) were ineffective for the desired hydrosilylation (Table 1, entries 3−5). Among other bidentate-phosphine ligands, BINAP resulted in a modest yield of 18% (Table 1, entry 6), while Xantphos showed a slightly better performance with a yield of 32% (Table 1, entry 7). Notably, [2-(diphenylphosphino)phenyl] ether (DPEphos) achieved a higher yield of 64% of the desired *gem*-difluorovinylsilane **3a** (Table 1, entry 8). For further improvement of the yield of this reaction, the effect of solvents including dichloromethane (DCM), hexane, THF, EtOAc, and 1,4-dioxane was evaluated. It was found that ether-based solvents showed better activity than other solvents, in which the best results were obtained with 1,4-dioxane leading to 98% of the NMR yield of the target product (Table 1, entries 9−13). Notably, the product revealed to be unstable during long purification over silica gel column chromatography, requiring rapid purification to achieve high yields. However, once isolated, it remains stable. Additionally, the evaluation of various palladium catalysts was conducted in this study, in which $Pd(OAc)_2$ showed the best catalytic performance, whereas other palladium catalysts such as $Pd_2(dba)_3{\cdot}CHCl_3$, $Pd(PPh_3)_4$, and $PdBr_2$ exhibited significantly lower or no activity (Table 1, entries 14−16). Overall, it was demonstrated by the optimization of the catalyst loading that the reaction could be achieved with only 2 mol % of $Pd(OAc)_2$ and 3 mol % of DPEphos (Table 1, entry 17).

The substrate scope of *gem*-difluoroallenes (**1a**) with various silanes (**2a−k**) was then investigated under the optimized reaction conditions (Scheme 2). The results showed that a range of diarylsilanes efficiently reacted with *gem*-difluoroallenes in the current system, exhibiting excellent chemo- and regioselectivity and yielding products **3** in moderate to excellent yields. The scope was composed of diarylsilanes with neutral substituents (**2a**), electron-donating substituents (**2b−e**), and a halogen atom (**2f**) leading to corresponding products **3a−f** with yields ranging from 56 to 90%. Arylalkylsilanes with different alkyl groups, such as methyl (**2g**), CH_2TMS (**2h**), and *tert*-butyl (**2i**, **2j**), could also afford the target products **3g−j** in high yields, indicating that increased steric hindrance showed no significant effects on the reaction yield and *β*-selectivity. Notably, the relatively lower isolated yields observed for certain arylalkylsilanes (**2g−2j**) are likely due to the instability of the corresponding hydrosilylation products on silica gel, which can lead to partial desilylation during purification. In contrast, products derived from diarylsilanes exhibited higher stability under the same conditions, thus enabling better yields after chromatographic separation. This observation highlights the different behaviors of arylalkylsilanes and diarylsilanes in the reaction and purification processes. Additionally, phenylsilane **2k** also showed good activity within this reaction delivering the corresponding product **3k** in 52% yield. In addition, under conditions involving excess $PhSiH_3$, the bis-hydrosilylation product (**3l**) was successfully obtained, and the monohydrosilylation product (**3k**) was not detected. This experiment indicates distinct differences in reactivity among different hydrosilanes and suggests a sequential addition process.

Moreover, the scope was further expanded to *gem*-difluoroallenes (Scheme 3) containing different aryl groups (R = 4-Me, 2-F, 4-Cl, 4-Br, 3-CF_3). They exhibited good reactivity with diarylsilanes and provided the target products **3m−3q** with 69%−88% yields and excellent chemo- and

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Scheme 2. Scope of Hydrosilanes for the Catalytic Hydrosilylation of *gem*-Difluoroallene 1a[a,b]



[a]Reaction conditions: *gem*-difluoroallene **1a** (0.2 mmol, 1.0 equiv), silane (1.2 equiv, 0.24 mmol), $Pd(OAc)_2$ (2 mol %, 0.004 mmol), DPEphos (3 mol %, 0.006 mmol), 1,4-dioxane (2 mL) at room temperature under Ar for 12 h. [b]**1a** (2.5 equiv, 0.5 mmol), silane (1.0 equiv, 0.2 mmol), $Pd(OAc)_2$ (5 mol %, 0.01 mmol), DPEphos (10 mol %, 0.02 mmol).

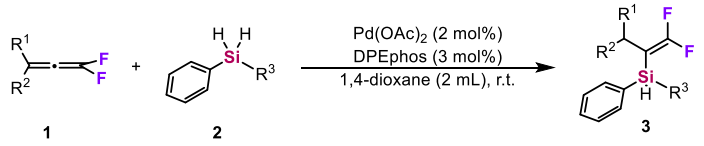
regioselectivity. Additionally, a substrate with −OBn substituents also showed good tolerance under these experimental conditions with phenyl *tert*-butylsilane furnishing the corresponding product **3t** in a fair 50% yield. Regarding the disubstituted aryl substrates (2,4-Cl_2, 3,4-Me_2), the reaction yielded **3r** and **3s** products with 52% and 82% yields, respectively, and the more polar substituents (−OMe, −CO_2Me) were also well-tolerated leading to the products **3u** and **3v** in 76% and 79% yields, respectively.

gem-Difluoroallenes containing heterocycles (furan, thiophene, 2,3-dioxane) exhibited good to excellent reactivity under these reaction conditions leading to the corresponding products **3w**−**3y** in yields ranging from 71 to 92%. Furthermore, substrates containing chlorine, *tert*-butyldiphenylsilyl (TBDP) ether terminal alkenes, and long alkyl chains also reacted successfully furnishing the corresponding products **3z**−**3ac**, with yields ranging from 62% to 74%. Notably, tetrasubstituted *gem*-difluoroallene also showed high efficiency in the reaction, with the target product **3ad** obtained in 81% yield, albeit without diastereoselectivity (*dr* value of 1:1). The broad substrate compatibility and excellent regioselectivity were demonstrated in this reaction system based on the above results, which could effectively accommodate various silanes and *gem*-difluoroallenes, and it could provide an efficient synthetic strategy for the construction of functionalized silanes and complex organic molecules. Additionally, these findings could further highlight the potential applications of this reaction and provide valuable insights into the development of *gem*-difluorovinyl chemistry. To further demonstrate the viability of our reaction, *gem*-difluoroallene from citronellal was successfully employed in this study, and the product **3ae** was obtained in 92% yield with complete regioselectivity.

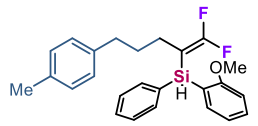
Gram-scale reactions of substrate **1a** were successfully conducted with diphenylsilane **2a** and phenylsilane **2k** and provided 1.12 g of product **3a** (60% yield) and 0.87 g of product **3k** (60% yield) with maintained high chemo- and regioselectivity. We further employed both products to demonstrate the versatile transformation potential of the *gem*-difluorovinylsilanes. From **3a**, hydrogenation under Pd/C in THF/H_2O as a solvent afforded silanol **4** in 90% yield, while methanol as a solvent afforded silyl ether **5** in 92% yield. Additionally, treatment with CH_2I_2/Et_2Zn produced alkyl-substituted silane **6** in 70% yield. The hydrosilylation of diphenylacetylene under $Pd(OAc)_2$/PPh_3 catalysis with **3k** afforded the vinylation product **7** in 72% yield. In addition, cyclization of **3k** with 1,4-butanediol under Cs_2CO_3 catalysis formed cyclic dioxasilolane **8** in 75% yield (Scheme 4A). Critical insights were revealed by mechanistic investigation in

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

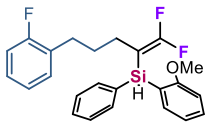
Scheme 3. Scope of *gem*-Difluoroallenes for the Catalytic Hydrosilylation[a,b]



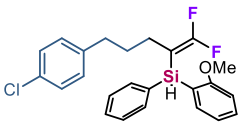
gem-Difluoroallenes Scope



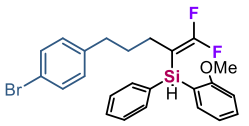
3m, 88%



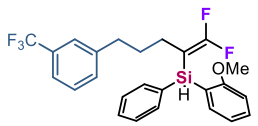
3n, 78%



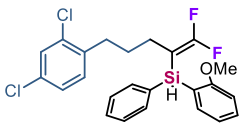
3o, 80%



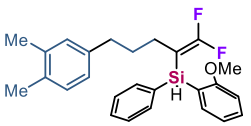
3p, 87%



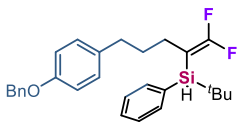
3q, 69%



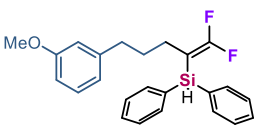
3r, 52%



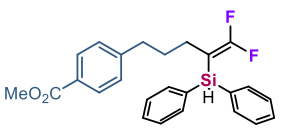
3s, 82%



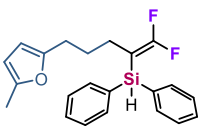
3t, 50%



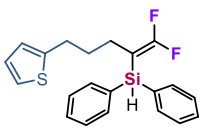
3u, 76%[b]



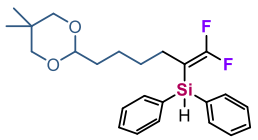
3v, 79%



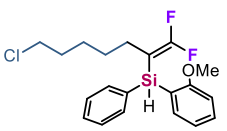
3w, 92%



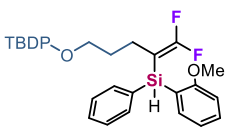
3x, 87%



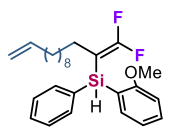
3y, 71%



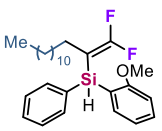
3z, 72%



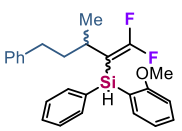
3aa, 74%



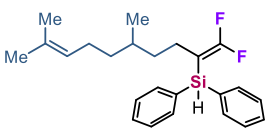
3ab, 62%



3ac, 68%



3ad, 81%, dr =1:1



3ae, 92% yield, rr > 99:1

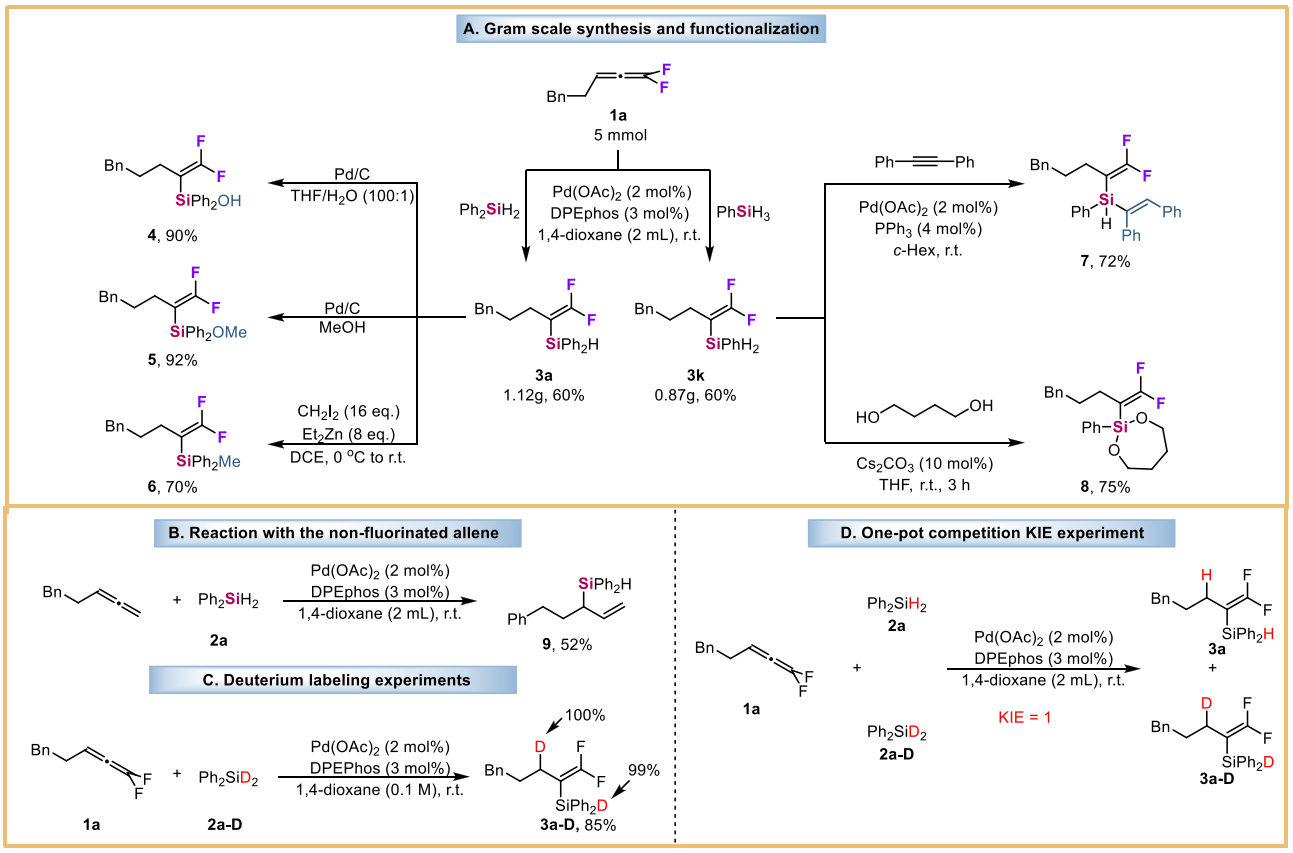
[a]Reaction conditions: *gem*-difluoroallene (0.2 mmol, 1.0 equiv), silane (1.2 equiv, 0.24 mmol), $Pd(OAc)_2$ (2 mol %, 0.004 mmol), DPEphos (3 mol %, 0.006 mmol), 1,4-dioxane (2 mL) at room temperature under Ar for 12 h. [b]Time = 36 h.

this study. The comparison between branched product **9** from nonfluorinated allene and our difluoroallene products **3** could suggest that the fluorine's electron-withdrawing effect governed the regiocontrol of the reaction (Scheme 4B). A deuterium labeling experiment with substrate **1a** and Ph_2SiD_2 (**2a**-D) demonstrated >99% of D-incorporation in vinylsilane **3a**-D (Scheme 4C), while kinetic isotope experiments (KIE = 1) seemed to exclude Si−H/D bond cleavage as the rate-determining step (Scheme 4D).

Building upon previous reports of Pd-catalyzed hydrosilylation[15a,18] and guided by our control experiments, we conducted a systematic mechanistic investigation using density functional theory (DFT) calculations with *gem*-difluoroallene (**1a**) and diphenylsilane (**2a**) as model substrates (Scheme 5). Combining experimental and computational insights, we elucidated the selective formation of major product **3a** in a Pd/DPEphos-catalyzed system. As shown in Scheme 5, the proposed catalytic cycle involves several key steps. Initially, the palladium precursor is reduced in situ by the phosphine ligand to generate the active Pd(0)Ln species (denoted as **PdL**), which serves as the reference point (ΔG = 0) for the Gibbs free energy profile. While palladium typically favors coordination with terminal alkenes, the electron-withdrawing effect of the fluorine substituents in **1a** shifts the preference to the internal C_2═C_3 double bond, forming intermediate **A**. DFT calculations confirmed that coordination at the distal double bond (C_2═C_3) is energetically favored by 11.8 kcal/mol over the proximal C_1═C_2 bond. Subsequent oxidative addition of

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Scheme 4. Control Experiments: (A) Gram-Scale Synthesis and Further Transformation; (B) Comparison with Nonfluorinated Allene; (C) Deuterium Labeling Experiments; (D) One-Pot Competition KIE Experiment



the Si−H bond in **2a** to **A** yields complex **B**, which undergoes migratory insertion of the olefin into the Pd−H bond, affording the silyl alkyl Pd(II) intermediate **C**. Four distinct transition states (**TS1-1**, **TS1-3**, **TS1-5**, and **TS1-7**) were located along pathways 1, 3, 5, and 7, respectively. This insertion step was identified as the rate-determining step (RDS), critically governing both the reactivity and regioselectivity. Comparative analysis of the two competing pathways, as H-insertion versus Si-insertion, revealed that the former consistently exhibits lower activation barriers (see Supporting Information). Finally, reductive elimination from **C** generates complex **D**, which releases the product and regenerates **PdL** to close the catalytic cycle. Energetically, the H-insertion pathway followed by reductive elimination proved to be the most favorable, with **TS1-1** exhibiting the lowest activation barrier (17.6 kcal/mol), consistent with the experimental predominance of **3a**. Intriguingly, despite minimal energy differences between **TS1-1** and **TS1-3**, the preferential coordination of Pd to the distal double bond (C_2=C_3) drives the selectivity toward **3a**. In contrast, **TS1-5** displayed the highest barrier (22.5 kcal/mol), attributed to the electron-withdrawing fluorine substituents, which render the γ-position less nucleophilic by inductively stabilizing a partial positive charge at the α-carbon. Notably, even in the absence of fluorine substituents (e.g., with hydrogen at the α-position), the H-insertion pathway remains energetically favored, though the product distribution shifts to the branched allylic silane **9** (see Supporting Information).

CONCLUSION

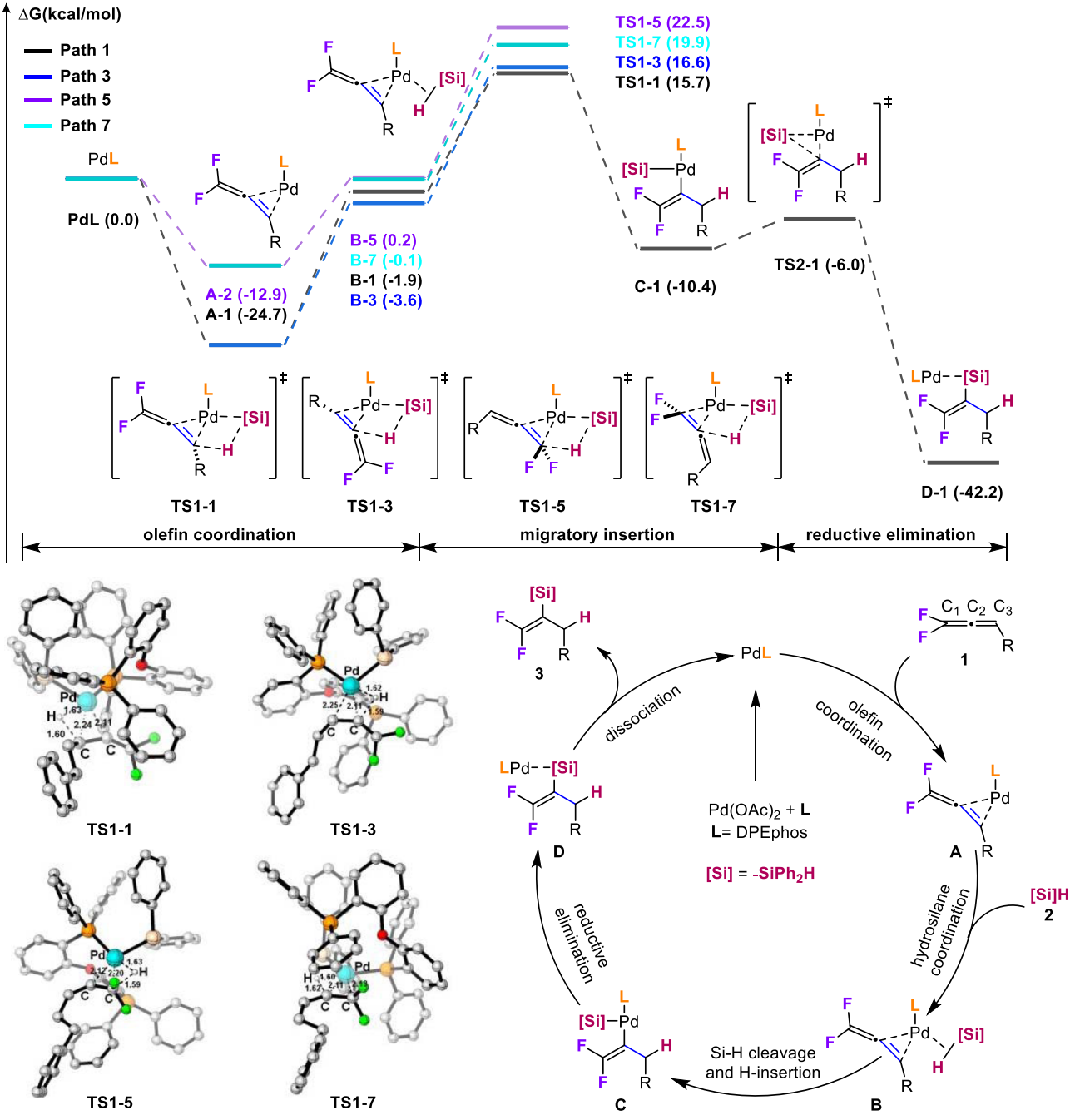
In summary, we have developed a novel palladium-catalyzed β-selective hydrosilylation of *gem*-difluoroallenes with hydrosilanes, providing efficient access to *gem*-difluorovinylsilanes. This method exhibits a broad substrate scope, delivering the target products in moderate to excellent yields, and demonstrates excellent compatibility with late-stage functionalization strategies. Mechanistic investigations through density functional theory (DFT) calculations revealed the key transition states, offering fundamental insights into the origin of the observed chemo- and regioselectivity. Given the mild reaction conditions and operational simplicity of this protocol, the resulting β-difluorovinylsilanes and their derivatives hold significant potential for diverse applications in synthetic chemistry and pharmaceutical research.

METHODS

General Procedure for the Synthesis of *gem*-Difluoroallenes 1. To a solution of $(i\text{-Pr})_2NH$ (20 mmol) in THF (10 mL) was added *n*-BuLi (20.0 mmol) over 10 min at 0 °C under argon. The resulting solution was allowed to stir for an additional 15 min and then cooled to −80 °C in a cold ethanol bath. To the cold LDA solution was added a solution of CF_3CH_2I (10.0 mmol) in THF (5 mL) over 10 min, keeping the temperature between −93 °C and −85 °C. After the mixture was stirred for 20 min at that same temperature, a solution of aldehyde (10.0 mmol) in THF (5 mL) was added over 5 min, keeping the temperature between −93 °C and −85 °C. The mixture was stirred for an additional 30 min, then warmed to −30 °C over 90 min. After Ac_2O (12.0 mmol) was

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Scheme 5. Mechanism of Palladium-Catalyzed Hydrosilylation of *gem*-Difluoroallenes[a]



[a]Free-energy profile (ΔG, in kcal·mol^{-1}) for Pd/DPEphos-catalyzed hydrosilylation of difluoroallene (**1a**) with diphenylsilane (**2a**) via four possible pathways (paths 1, 3, 5, and 7) calculated at the M06L/def2TZVP//M06L/SDD(Pd), 6-31G(d,p) (other elements) level of theory.

added, the mixture was allowed to warm to 0 °C over 2 h. The reaction was quenched with sat. aq NH_4Cl (20 mL), and the products were extracted with ethyl acetate (3 × 20 mL). The combined organic layer was washed with brine (20 mL) and dried (Na_2SO_4). After the solvent was removed under reduced pressure, the residue was purified by flash column chromatography giving acetate S1 (see Supporting Information). To a suspension of zinc powder (17.0 mmol) in DMF (26 mL) was added a solution of S1 (8.50 mmol) in DMF (17 mL) at room temperature under argon, and the mixture was stirred for 3 h. The resulting mixture was filtered to remove the excess zinc and then diluted with ethyl acetate (20 mL) and brine (15 mL). The products were extracted with ethyl acetate (3 × 15 mL). The combined organic layer was washed with brine (15 mL) and dried (Na_2SO_4). After the solvent was removed under reduced pressure, the residue was purified by flash column chromatography (pentane) to give compound **1**.

General Procedure for the Synthesis of *gem*-Difluorovinylsilanes 3. A Schlenk tube was purged with argon three times, and to the tube were added $Pd(OAc)_2$ (0.9 mg, 2 mol %), DPEphos (3.2 mg, 3 mol %), and 1,4-dioxane (2 mL) under an argon atmosphere. Then, 1,1-difluoroallene **1** (0.2 mmol, 1.0 equiv) and silane **2** (0.24 mmol, 1.2 equiv) were added subsequently. The reaction mixture was stirred at

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

room temperature for 12 h. The resulting mixture was concentrated under reduced pressure, and the residue was purified by flash column chromatography on silica gel to afford the difluorovinylsilanes **3**.

General Procedure for the Synthesis of (1,1-Difluoro-5-phenylpent-1-en-2-yl)diphenylsilanol 4. A solution of **3a** (72.8 mg, 0.2 mmol) in THF (2.0 mL, containing 1% v/v H_2O) was added to a flame-dried Schlenk tube under argon, and Pd/C (15.0 mg, 10 wt %) was added and stirred at room temperature. After the reaction was completed (10 min), DCM (5.0 mL) was added and dried by Na_2SO_4; then the reaction mixture was filtered through a pad of Celite, eluted with ethyl acetate, concentrated in vacuo, and purified by silica gel chromatography (pentane:EA = 100/0 to 95/5) to give the product **4**.

General Procedure for the Synthesis of (1,1-Difluoro-5-phenylpent-1-en-2-yl)(methoxy)diphenylsilane 5. A solution of **3a** (72.8 mg, 0.2 mmol) in MeOH (2.0 mL) was added to a flame-dried Schlenk tube under argon, Pd/C (15.0 mg, 10 wt %) was added, and the mixture was stirred at room temperature. After the reaction was completed, DCM (5.0 mL) was added and dried by Na_2SO_4; then the reaction mixture was filtered through a pad of Celite, eluted with ethyl acetate, concentrated in vacuo, and purified by silica gel chromatography (pentane:EA = 100/0 to 98/2) to give the product **5**.

General Procedure for the Synthesis of (1,1-Difluoro-5-phenylpent-1-en-2-yl)(methyl)diphenylsilane 6. A solution of **3a** (72.8 mg, 0.2 mmol) in DCE (2.0 mL) was added to a flame-dried Schlenk tube under argon; CH_2I_2 (16.0 eq, 3.2 mmol) and Et_2Zn (8.0 eq, 1.6 mmol) were added and stirred at room temperature. After the reaction was completed, DCM (5.0 mL) was added and dried by Na_2SO_4; then the reaction mixture was filtered through a pad of Celite, eluted with ethyl acetate, concentrated in vacuo, and purified by silica gel chromatography to give the product **6**.

General Procedure for the Synthesis of (*E*)-(1,1-Difluoro-5-phenylpent-1-en-2-yl)(1,2-diphenylvinyl)-(phenyl)silane 7. In a flame-dried Schlenk tube, were added $Pd(OAc)_2$ (0.9 mg, 2 mol %), PPh_3 (2.1 mg, 4 mol %) in cyclohexane (1 mL, 0.2 M). Then, **3k** (57.7 mg, 0.2 mmol) and diphenylacetylene (45.2 mg, 0.2 mmol) were added sequentially to the reaction mixture, and the reaction tube was cooled at room temperature and then stirred for 12 h. After completion of the reaction, the mixture was passed through a short Celite pad, using DCM as a solvent. The residue was purified by column chromatography on silica gel to afford the corresponding product **7** as a colorless oil.

General Procedure for the Synthesis of 2-(1,1-Difluoro-5-phenylpent-1-en-2-yl)-2-phenyl-1,3,2-dioxasilepane 8. Inside a nitrogen-atmosphere glovebox, Cs_2CO_3 (0.02 mmol, 10 mol %, 6.6 mg) was charged into an oven-dried 10 mL Schlenk tube equipped with a stirring bar. The reaction tube was capped tightly and removed from the glovebox. Subsequently, silane **3k** (0.2 mmol, 1.0 equiv, 57.7 mg) and 1,4-butanediol (0.24 mmol, 1.2 equiv, 43.2 mg) and THF (3.0 mL) were added under an argon atmosphere and then stirred at room temperature for 3 h. Then the mixture was diluted and extracted with DCM. The organic layer was washed with brine, dried with anhydrous Na_2SO_4, filtered, and concentrated in vacuo. The residue was purified by column chromatography on silica gel to afford the corresponding product **8** as a colorless oil.

General Procedure for the Synthesis of Diphenyl(5-phenylpent-1-en-3-yl)silane 9. The Schlenk tube was purged with argon three times, and to the tube were added $Pd(OAc)_2$ (0.9 mg, 2 mol %), DPEphos (3.2 mg, 3 mol %), and 1,4-dioxane (2 mL) under an argon atmosphere. Then, allene (0.2 mmol, 1.0 equiv) and Ph_2SiH_2 (0.24 mmol, 1.2 equiv) were added subsequently. The reaction mixture was stirred at room temperature for 12 h. The resulting mixture was concentrated under reduced pressure, and the residue was purified by flash column chromatography on silica gel to afford the difluorovinylsilanes **9**.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c03553.

Reaction optimization, experimental procedures, characterization data, and copies of ^1H, ^{13}C, and ^{19}F spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Samuel Couve-Bonnaire − *Univ Rouen Normandie, INSA Rouen Normandie, Univ Caen Normandie, ENSICAEN, CNRS, Institut CARMeN UMR 6064, F-76000 Rouen, France*; orcid.org/0000-0003-2149-0849; Email: samuel.couve-bonnaire@insa-rouen.fr

Li-Wen Xu − *Key Laboratory of Organosilicon Chemistry and Material Technology of Ministry of Education, and Zhejiang Key Laboratory of Organosilicon Material Technology, College of Material, Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 311121, P. R. China*; orcid.org/0000-0001-5705-0015; Email: liwenxu@hznu.edu.cn

Jean-Philippe Bouillon − *Univ Rouen Normandie, INSA Rouen Normandie, Univ Caen Normandie, ENSICAEN, CNRS, Institut CARMeN UMR 6064, F-76000 Rouen, France*; orcid.org/0000-0002-6274-1125; Email: jean-philippe.bouillon@univ-rouen.fr

Authors

Xin-Long Yu − *Univ Rouen Normandie, INSA Rouen Normandie, Univ Caen Normandie, ENSICAEN, CNRS, Institut CARMeN UMR 6064, F-76000 Rouen, France*

An-Jiu Wen − *Key Laboratory of Organosilicon Chemistry and Material Technology of Ministry of Education, and Zhejiang Key Laboratory of Organosilicon Material Technology, College of Material, Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 311121, P. R. China*

Zheng Xu − *Key Laboratory of Organosilicon Chemistry and Material Technology of Ministry of Education, and Zhejiang Key Laboratory of Organosilicon Material Technology, College of Material, Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 311121, P. R. China*; orcid.org/0000-0003-4704-0846

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.5c03553

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Author Contributions

The manuscript was written through contributions of all authors. All the authors have given approval to the final version of the manuscript.

Funding

This work was supported by the Normandie Université (NU), the Région Normandie, the Centre National de la Recherche Scientifique (CNRS), Université de Rouen Normandie (URN), INSA Rouen Normandie, Labex SynOrg (ANR-11-LABX-0029), the Graduate School for Research XL-Chem (ANR-18-EURE-0020 XL CHEM), and Innovation Chimie Carnot (I2C). National Natural Science Foundation of China (No. 22371060 and 22361162606), Zhejiang Provincial Natural Science Foundation of China (No. LZ23B020002), and Interdisciplinary Research Project of Hangzhou Normal University (2025JCXK02) are gratefully acknowledged.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

X.-L.Y. thanks the Région Normandie (RIN 50% program) for a doctoral fellowship. The authors also thank Dr. J. Cao and Dr. J.H. Ma for their assistance in the spectral analysis.

■ REFERENCES

(1) (a) Selected reviews on organofluorine chemistry: (b) Purser, S.; Moore, P. R.; Swallow, S.; Gouverneur, V. Fluorine in medicinal chemistry. *Chem. Soc. Rev.* **2008**, *37*, 320−330. (c) Furuya, T.; Kamlet, A. S.; Ritter, T. Catalysis for fluorination and trifluoromethylation. *Nature* **2011**, *473*, 470−477. (d) Wang, J.; Sánchez-Roselló, M.; Aceña, J. L.; del Pozo, C.; Sorochinsky, A. E.; Fustero, S.; Soloshonok, V. A.; Liu, H. Fluorine in Pharmaceutical Industry: Fluorine-Containing Drugs Introduced to the Market in the Last Decade (2001−2011). *Chem. Rev.* **2014**, *114*, 2432−2506. (e) Pons, A.; Delion, L.; Poisson, T.; Charette, A. B.; Jubault, P. Asymmetric Synthesis of Fluoro, Fluoromethyl, Difluoromethyl, and Trifluoromethylcyclopropanes. *Acc. Chem. Res.* **2021**, *54*, 2969−2990. (f) Xiao, P.; Pannecoucke, X.; Bouillon, J. P.; Couve-Bonnaire, S. Wonderful fusion of organofluorine chemistry and decarboxylation strategy. *Chem. Soc. Rev.* **2021**, *50*, 6094−6151. (g) Roy, S.; Besset, T. New Opportunities to Access Fluorinated Molecules Using Organophotoredox Catalysis via C(sp^3)−F Bond Cleavage. *JACS Au* **2025**, *5*, 466−485.

(2) (a) Hagmann, W. K. The Many Roles for Fluorine in Medicinal Chemistry. *J. Med. Chem.* **2008**, *51*, 4359−4369. (b) Zhou, Y.; Wang, J.; Gu, Z.; Wang, S.; Zhu, W.; Acena, J. L.; Soloshonok, V. A.; Izawa, K.; Liu, H. Next Generation of Fluorine-Containing Pharmaceuticals, Compounds Currently in Phase II−III Clinical Trials of Major Pharmaceutical Companies: New Structural Trends and Therapeutic Areas. *Chem. Rev.* **2016**, *116*, 422−518.

(3) (a) Pan, Y.; Qiu, J.; Silverman, R. B. Design, Synthesis, and Biological Activity of a Difluoro-Substituted, Conformationally Rigid Vigabatrin Analogue as a Potent γ-Aminobutyric Acid Aminotransferase Inhibitor. *J. Med. Chem.* **2003**, *46*, 5292−5293. (b) Altenburger, J.-M.; Lassalle, G. Y.; Matrouaui, M.; Galtier, D.; Jetha, J.; Bocskei, Z.; Berry, C. N.; Lunven, C.; Lorrain, J.; Herault, J.; Schaeffer, P.; O'Commor, S. E.; Herbert, J. SSR182289A, a selective and potent orally active thrombin inhibitor. *Bioorg. Med. Chem.* **2004**, *12*, 1713−1730.

(4) (a) Meanwell, N. A. Synopsis of Some Recent Tactical Application of Bioisosteres in Drug Design. *J. Med. Chem.* **2011**, *54*, 2529−2591. (b) Magueur, G.; Crousse, B.; Ourévitch, M. Fluoroartemisinins: When a *gem*-difluoroethylene replaces a carbonyl group. *J. Fluorine Chem.* **2006**, *127*, 637−642. (c) Leriche, C.; He, X.; Chang, C.-W. T.; Liu, H.-W. Reversal of the Apparent Regiospecificity of NAD(P)H-Dependent Hydride Transfer: The Properties of the Difluoromethylene Group, A Carbonyl Mimic. *J. Am. Chem. Soc.* **2003**, *125*, 6348−6349. (d) Zhao, Z.; Liu, H.-W. Synthesis of a Deoxysugar Dinucleotide Containing an exo-Difluoromethylene Moiety as a Mechanistic Probe for Studying Enzymes Involved in Unusual Sugar Biosynthesis. *J. Org. Chem.* **2001**, *66*, 6810−6815.

(5) (a) Fujita, T.; Fuchibe, K.; Ichikawa, J. Transition-Metal-Mediated and -Catalyzed C-F Bond Activation by Fluorine Elimination. *Angew. Chem., Int. Ed.* **2019**, *58*, 390−402. (b) Hu, J.; Han, X.; Yuan, Y.; Shi, Z. Stereoselective Synthesis of *Z* Fluoroalkenes through Copper Catalyzed Hydrodefluorination of *gem*-Difluoroalkenes with Water. *Angew. Chem., Int. Ed.* **2017**, *56*, 13342−13346. (c) Yao, C.; Wang, S.; Norton, J.; Hammond, M. Catalyzing the Hydrodefluorination of CF_3-Substituted Alkenes by $PhSiH_3$. H• Transfer from a Nickel Hydride. *J. Am. Chem. Soc.* **2020**, *142*, 4793−4799.

(6) (a) Krespan, C. G.; Petrov, V. A. The Chemistry of Highly Fluorinated Carbocations. *Chem. Rev.* **1996**, *96*, 3269−3301. (b) Olah, G. A.; Mo, Y. K. Organic Fluorine Compounds. XXXIII. Electrophilic Additions to Fluoro Olefins in Superacids. *J. Org. Chem.* **1972**, *37*, 1028−1034.

(7) Sathish, K.; Jain, S.; Sihag, N.; Yadav, M. R. *gem*-Difluoroallene (DFA): an emerging versatile synthetic precursor to enable diverse fluorinated molecules. *Org. Biomol. Chem.* **2024**, *22*, 8078−8096.

(8) Han, X.; Wang, M.; Liang, Y.; Zhao, Y.; Shi, Z. Regio- and enantioselective nucleophilic addition to *gem*-difluoroallenes. *Nat. Synth.* **2022**, *1*, 227−234.

(9) (a) Wang, C. Q.; Li, Z. Q.; Tian, L.; Walsh, P. J.; Feng, C. Electronically controlled regioselective hydroarylation of gem-difluoroallenes. *Cell Rep. Phys. Sci.* **2022**, *3*, 101117−101131. (b) Li, Z. Q.; Tang, H. J.; Wang, Z.; Wang, C. Q.; Feng, C. Multi-substituted trifluoromethyl alkene construction via gold-catalyzed fluoroarylation of *gem*-difluoroallenes. *Chem. Sci.* **2024**, *15*, 3524−3529. (c) Li, W.; Wang, C.; Xiao, M.; Cheng, L. J. Copper-Catalyzed Protoarylation of *gem*-Difluoroallenes. *Org. Lett.* **2024**, *26*, 525−529.

(10) Li, Y.-Y.; Gao, B. Rhodium-Catalyzed Enantioselective *N*-Allylation of Sulfoximines. *Org. Lett.* **2023**, *25*, 2756−2760.

(11) Fuchibe, K.; Abe, M.; Sasaki, M.; Ichikawa, J. Gold-catalyzed electrophilic activation of 1,1-difluoroallenes: α- and γ-selective addition of heteroatom nucleophiles. *J. Fluorine Chem.* **2020**, *232*, 109452.

(12) Ji, D.; Qi, Z.; Li, X. Palladium-Catalyzed Regio- and Enantioselective Hydrophosphination of *gem*-Difluoroallenes. *Org. Lett.* **2023**, *25*, 5957−5962.

(13) (a) Wang, C. Q.; Li, Y.; Feng, C. Sterically congested boronate and silane synthesis via electronically controlled protoboration and protosilylation. *Cell Rep. Phys. Sci.* **2021**, *2*, 100461−100474. (b) Shan, C. C.; Dai, K. Y.; Zhao, M.; Xu, Y. H. Copper Catalyzed Protosilylation/Protoborylation of *gem*-Difluoroallenes. *Eur. J. Org Chem.* **2021**, 4054−4058.

(14) Li, L.; Huang, W. S.; Xu, Z.; Xu, L. W. Catalytic asymmetric silicon-carbon bond-forming transformations based on Si-H functionalization. *Sci. China: Chem.* **2023**, *66*, 1654−1687.

(15) (a) Yang, X.; Yuan, C.; Ge, S. Ligand-enabled stereodivergence in nickel-catalyzed regioselective hydroboration of internal allenes. *Chem* **2023**, *9*, 198−215. (b) Su, W.; Zhu, J.; Chen, Y.; Zhang, X.; Qiu, W.; Yang, K.; Yu, P.; Song, Q. Copper-catalysed asymmetric hydroboration of alkenes with 1,2-benzazaborines to access chiral naphthalene isosteres. *Nat. Chem.* **2024**, *16*, 1312−1319. (c) Li, X.; Song, Q. Asymmetric synthesis of chiral sp3-geminated borylstannyl, borylsilyl, and diboryl compounds. *Org. Chem. Front.* **2025**, *12*, 3129−3142.

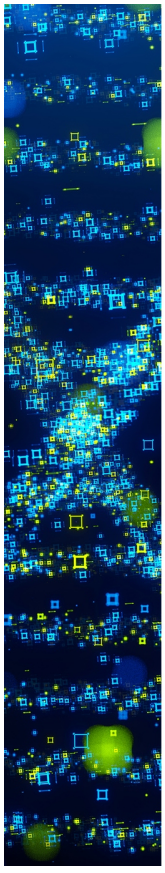
(16) (a) Miller, Z. D.; Dorel, R.; Montgomery, J. Regiodivergent and Stereoselective Hydrosilylation of 1,3-Disubstituted Allenes. *Angew. Chem., Int. Ed.* **2015**, *54*, 9088−9091. (b) Zeng, J. H.; Chen, J. J.; Chen, L.; Zhan, Z. P. Access to branched allylsilanes by nickel-catalyzed regioselective hydrosilylation of allenes. *Org. Chem. Front.* **2020**, *7*, 1132−1136.

(17) (a) Miller, Z. D.; Li, W.; Belderrain, T. R.; Montgomery, J. Regioselective Allene Hydrosilylation Catalyzed by N-Heterocyclic

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756

Carbene Complexes of Nickel and Palladium. *J. Am. Chem. Soc.* **2013**, *135*, 15282−15285. (b) Xie, H.; Zhao, L.; Yang, L.; Lei, Q.; Fang, W.; Xiong, C. Mechanisms and Origins of Switchable Regioselectivity of Palladium- and Nickel-Catalyzed Allene Hydrosilylation with N-Heterocyclic Carbene Ligands: A Theoretical Study. *J. Org. Chem.* **2014**, *79*, 4517−4527. (c) Miller, Z. D.; Montgomery, J. Regioselective Allene Hydroarylation via One-Pot Allene Hydrosilylation/Pd-Catalyzed Cross-Coupling. *Org. Lett.* **2014**, *16*, 5486−5489. (d) Tafazolian, H.; Schmidt, J. A. R. Highly efficient regioselective hydrosilylation of allenes using a [(3IP)Pd(allyl)]OTf catalyst; first example of allene hydrosilylation with phenyl- and diphenylsilane. *Chem. Commun.* **2015**, *51*, 5943−5946. (e) Chen, J. J.; Zeng, J. H.; Yang, Y.; Liu, Z. K.; Jiang, Y. N.; Li, M. R.; Chen, L.; Zhan, Z. P. A Bithiophene-Promoted ppm Levels of Palladium-Catalyzed Regioselective Hydrosilylation of Terminal Allenes. *Adv. Synth. Catal.* **2020**, *362*, 2360−2366. (f) Sudo, T.; Asao, N.; Gevorgyan, V.; Yamamoto, Y. Lewis Acid Catalyzed Highly Regio- and Stereocontrolled Trans-Hydrosilylation of Alkynes and Allenes. *J. Org. Chem.* **1999**, *64*, 2494−2499. (g) Asako, S.; Ishikawa, S.; Takai, K. Synthesis of Linear Allylsilanes via Molybdenum-Catalyzed Regioselective Hydrosilylation of Allenes. *ACS Catal.* **2016**, *6*, 3387−3395. (h) Yang, Z.; Peng, D.; Du, X.; Huang, Z.; Ma, S. Identifying a cobalt catalyst for highly selective hydrosilylation of allenes. *Org. Chem. Front.* **2017**, *4*, 1829−1832. (i) Wang, C.; Teo, W. J.; Ge, S. Access to stereodefined (*Z*)-allylsilanes and (*Z*)-allylic alcohols via cobalt-catalyzed regioselective hydrosilylation of allenes. *Nat. Commun.* **2017**, *8*, 2258. (j) Li, K.; Nie, M.; Tang, W. Synthesis of α-tertiary allylsilanes by palladium-catalyzed hydrosilylation of 1,1-disubstituted allenes. *Green Synth. Catal.* **2020**, *1*, 171−174. (k) Chen, S.; He, X.; Jin, Y.; Lan, Y.; Shen, X. Copper-catalyzed regio- and stereo-selective hydrosilylation of terminal allenes to access (*E*)-allylsilanes. *Nat. Commun.* **2022**, *13*, 3691. (l) Xu, J. L.; Xu, Z. Y.; Wang, Z. L.; Ma, W. W.; Sun, X. Y.; Fu, Y.; Xu, Y. H. Copper-Catalyzed Regiodivergent and Enantioselective Hydrosilylation of Allenes. *J. Am. Chem. Soc.* **2022**, *144*, 5535−5542. (m) Li, S.; Xu, J. L.; Xu, Y. H. Copper-Catalyzed Enantioselective Hydrosilylation of Allenes to Access Axially Chiral (Cyclohexylidene)ethyl Silanes. *Org. Lett.* **2022**, *24*, 6054−6059. (n) Liu, T.; Mao, X. R.; Song, S.; Chen, Z. Y.; Wu, Y.; Xu, L. P.; Wang, P. Enantioselective Nickel-Catalyzed Hydrosilylation of 1,1-Disubstituted Allenes. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202216878. (o) Guo, J.; Cheng, Z.; Chen, J.; Chen, X.; Lu, Z. Iron- and Cobalt-Catalyzed Asymmetric Hydrofunctionalization of Alkenes and Alkynes. *Acc. Chem. Res.* **2021**, *54*, 2701−2716. (p) Sun, Y.; Wang, B.; Long, L.; Lou, J. A.; Chen, H.; Lu, Z. Ligand Relay Cobalt Catalysis for Vicinal Si,O-Alkanes from Terminal Alkynes. *ACS Catal.* **2024**, *14*, 8405−8413. (q) Chen, J.; Wei, W. T.; Li, Z.; Lu, Z. Metal-catalyzed Markovnikov-type selective hydrofunctionalization of terminal alkynes. *Chem. Soc. Rev.* **2024**, *53*, 7566−7589. (r) Kim, J. A.; Kim, S.; Tambe, S. D.; Jang, J.; Cho, E. J. Nickel-catalyzed stereocontrolled 2,3-hydrosilylation of 1,1-disubstituted allenes. *Chem. Sci.* **2025**, *16*, 7489−7494.

(18) (a) Xie, J. L.; Xu, Z.; Zhou, H. Q.; Nie, Y. X.; Cao, J.; Yin, G. W.; Bouillon, J.-P.; Xu, L. W. Palladium-catalyzed hydrosilylation of ynones to access siliconstereogenic silylenones by stereospecific aromatic interactionassisted Si−H activation. *Sci. China: Chem.* **2021**, *64*, 761−769. (b) Ling, F. Y.; Ye, F.; Fang, X. J.; Zhou, X. H.; Huang, W. S.; Xu, Z.; Xu, L. W. An unusual autocatalysis with an air-stable Pd complex to promote enantioselective synthesis of Si-stereogenic enynes. *Chem. Sci.* **2023**, *14*, 1123−1131. (c) Zhou, H. Q.; Ling, F. Y.; Fang, X. J.; Zhu, H. J.; Li, L.; Ye, F.; Xu, Z.; Xu, L. W. An unusual *trans*-hydrosilylation of prochiral 1,1-disubstituted cyclopropenes revealing the different nature of asymmetric palladium and rhodium catalysis. *Org. Chem. Front.* **2023**, *10*, 430−439. (d) Xu, H.; Yu, H. X.; Wen, A. J.; Ling, F. Y.; Ye, F.; Li, L.; Kwong, F. Y.; Xu, L. W. General ppm-level Pd-catalysed asymmetric diarylalkyne hydrosilylation to access structurally diverse Si-stereogenic vinylsilanes. *Org. Chem. Front.* **2024**, *12*, 106−114.



CAS BIOFINDER DISCOVERY PLATFORM™

STOP DIGGING THROUGH DATA —START MAKING DISCOVERIES

CAS BioFinder helps you find the right biological insights in seconds

Start your search

CAS
A division of the American Chemical Society

https://doi.org/10.1021/acscatal.5c03553
ACS Catal. 2025, 15, 13747−13756